QNSMARTI:H:QPCODE Reg.No : ..... Name :

.....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) QNSMARTI:H:EXCAPSTYPE EXAMINATION QNSMARTI:H:Y:MONTH QNSMARTI:H:Y:YEAR

QNSMARTI:H:S:SEMLETTER SEMESTER

QNSMARTI:H:C:COURSENATURE - QNSMARTI:H:C:CODE - QNSMARTI:H:C:CAPSNAME

(QNSMARTI:H:C:REVYEAR ADMISSION ONWARDS)

Duration: 1hr 30min **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short answer Questions

Answer any **5** questions

Each question carries **2** marks

1. Write any two **laboratory safety rules**. [U / CO1]
2. Define **aseptic technique**. [U / CO1]
3. What is meant by **pure culture**? [U / CO3]
4. Define **DNA extraction**. [U / CO4]
5. What is the role of **agar** in culture media? [U / CO3]
6. What is **PCR**? [U / CO5]
7. Name any two **types of staining techniques**. [K / CO3]

[2x5 = 10]

Part B

Short Essay Questions

Answer any 4 questions

Each question carries **5** marks

8. Explain the importance of **laboratory safety and biosafety measures**. [U / CO1]
9. Describe the **general characteristics of microorganisms**. [An / CO2]
10. Explain different **culture media** used for bacterial growth. [U / CO3]
11. Describe the **steps involved in isolation of bacteria**. [A / CO3]
12. Explain the **principle and steps of DNA extraction**. [A / CO4]
13. Describe the **principle and applications of PCR**. [A / CO5]

[5x4 = 20]

Part C

Long Essay Questions

Answer any **2** questions

Each question carries **10** marks

14. Discuss **laboratory safety measures, sterilization methods, and waste disposal practices**. [An / CO1]
15. Explain the **morphological and cultural characteristics of bacteria** with suitable examples. [An / CO2]
16. Describe in detail the **DNA extraction and quantification techniques**. [An / CO4]
17. Explain the **principle, procedure, and applications of PCR**. [An / CO5]

[10x2 = 20]